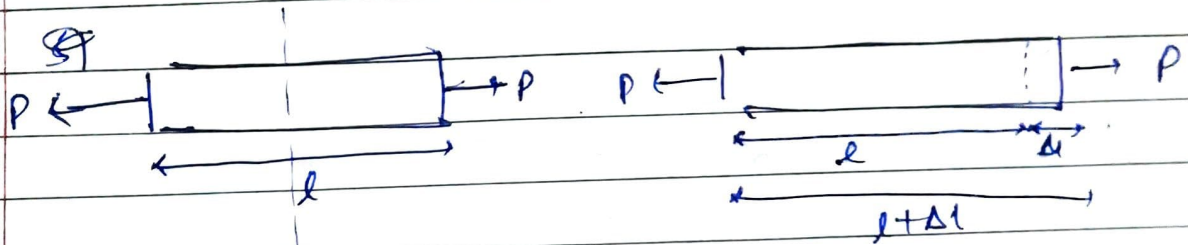


Unit - 2

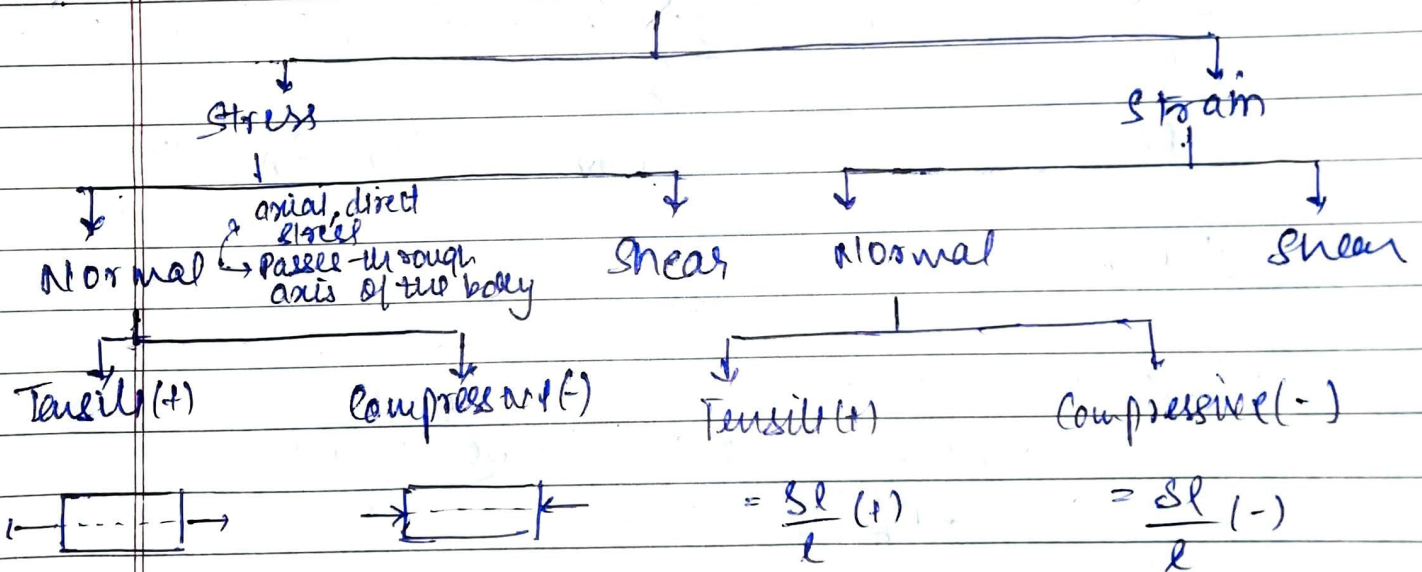
Introduction to Strength of Materials

Stress and Strain



$$\text{Stress} = \frac{\text{Resisting Force}}{\text{Cross sectional Area}}$$

Stress and Strain

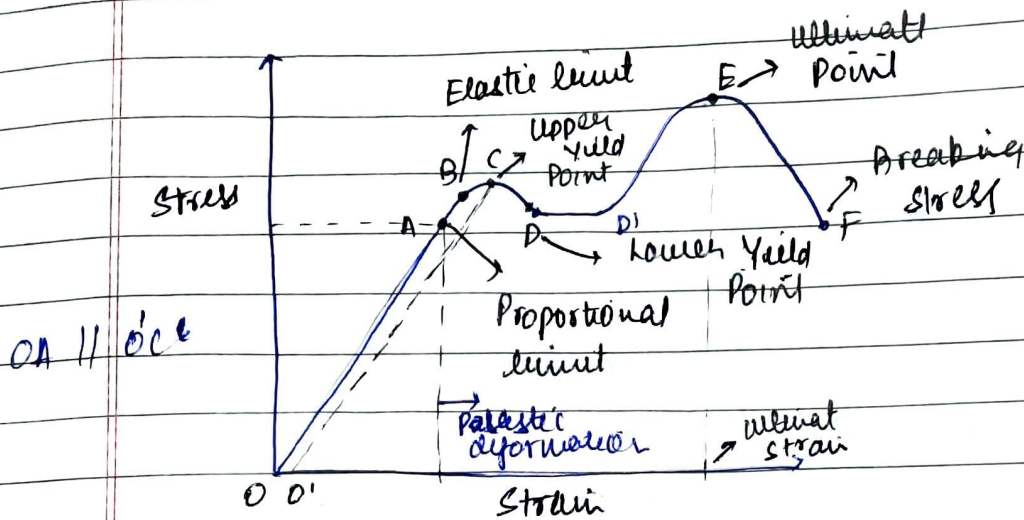


$$\sigma = \frac{P}{A}$$

10^6 MegaPascal
 10^9 GPa

$$\epsilon = \frac{\text{deformation}}{\text{original}}$$

Stress / Strain Curve



$O \rightarrow A \rightarrow$ Hook's Law valid

$$\sigma \propto \epsilon$$

$$\sigma = E \epsilon$$

[E = Bulk's Modulus, modulus of elasticity]

↳ slope

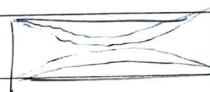
B $\rightarrow$ Elastic limit $\rightarrow$ The path in which it got elongated in the same path it'll follow to regain its shape.

B $\rightarrow$ C $\rightarrow$ Plastic deformation $\rightarrow$ If deformed back it'll follow another path CO'

D $\rightarrow$ D' $\rightarrow$ yielding $\rightarrow$ no increase in stress

E $\rightarrow$ maximum stress

Beyond E $\rightarrow$ necking



it decrease in cross-section area.

D' - E $\rightarrow$ strain hardening

F $\rightarrow$ fracture / breaking point

$$\text{Stiffness} = K = \frac{P}{\text{Deformation}}$$

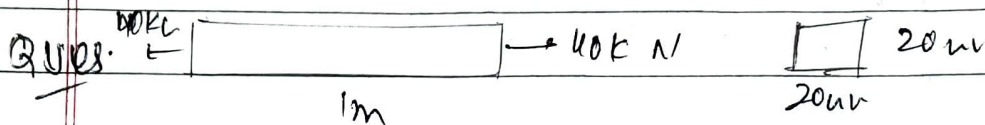
↳ resistance to deformation

$\rightarrow$ It's a load
Deformation $\rightarrow$ It's a deformation

Now for Hook's law

$$\sigma = E \epsilon$$

$$\frac{P}{A} = E \frac{\Delta l}{l} \Rightarrow \boxed{\Delta l = \frac{Pl}{AE}}$$



$$E = 200 \text{ GPa}$$

$$\Delta l = \frac{Pl}{AE} = \frac{40 \times 10^3 \times 1}{(20)^2 \times 10^{-6} \times 200 \times 10^9}$$

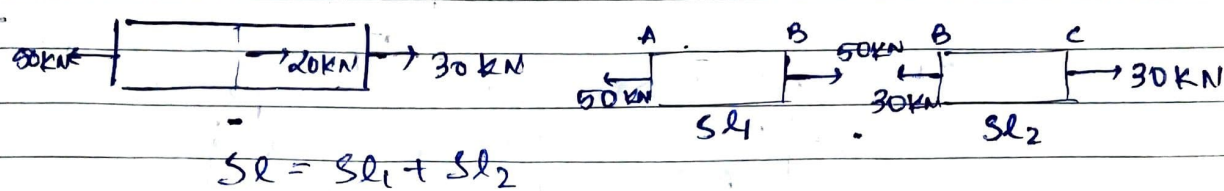
$$\Delta l = 5 \times 10^{-4} \text{ m}$$

$$\Delta l = 0.5 \text{ mm}$$

Principle of superposition

50 ← [] → 20 → 30 kN An axial load & $\sum F_x = 0$ hence it is in equilibrium

50 ← [] → x → 30 $\sum F_x = 0 \Rightarrow -50 + (x + 30) = 0$
 $x = 20$



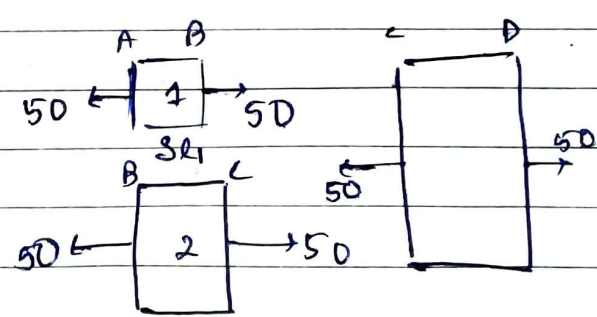
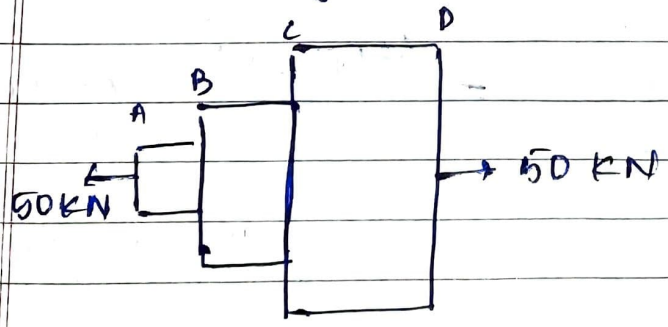
A B C D
 50 ← [] → 20 → 30 → x $\sum F_x = 0$ $x = 60$

From left :-
 50 ← [] → 50 30 ← [] → 30 50 ← [] → 60

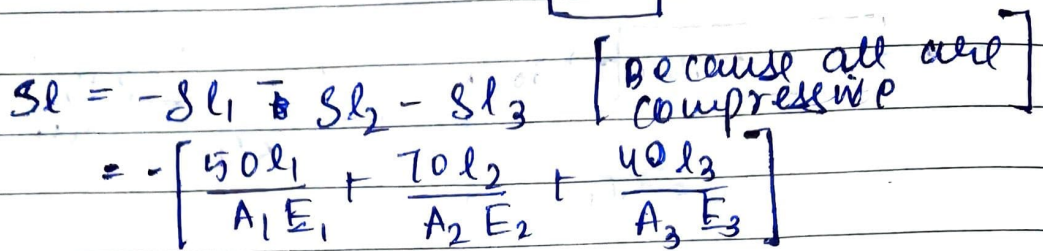
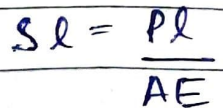
From right :-
 50 ← [①] → 50 30 ← [②] → 30 60 ← [③] → 60

$Sl = Sl_1 + Sl_2 + Sl_3$

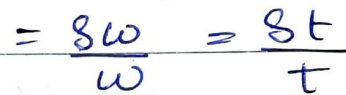
If anywhere the force is compressive then instead of + Sl_n it becomes - Sl_n



If axial load is applied at the ends then it will produce equal effect on all parts of the cross section areas.

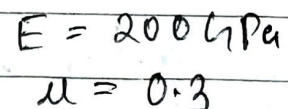



 longitudinal strain $\Rightarrow \frac{\Delta l}{l}$
 lateral strain $= -\frac{\Delta d}{d}$



only applicable upto elastic limit

→ Poisson's ratio helps in finding lateral strain i.e. $\epsilon_{\text{Lateral}}$ & when we can find σ_d

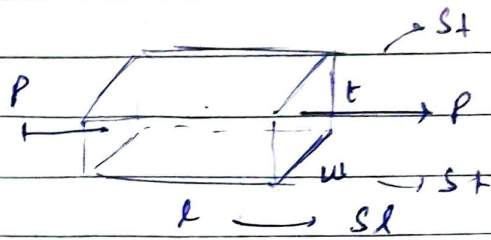


$$\frac{3l}{AE} = \frac{Pl}{AE} \Rightarrow e_{ln} = \frac{Sl}{l} \Rightarrow e_{lt} = 11 e_{ln} = 0.3 e_{ln}$$

$$e_{lt} = \frac{Sl}{t}$$

Volumetric Strain

$$\epsilon_v = \frac{\Delta V}{V}$$



$$\Delta V = V_f - V_i$$

$$= (l + sl)(w + sw)(t + st) - lwt$$

$$\frac{\Delta V}{V} = \frac{sw + sl + wst + l + sl}{lwt}$$

$$\left[\epsilon_v = \frac{sl}{l} + \frac{st}{t} + \frac{sw}{w} \right] = \epsilon_l + \epsilon_t + \epsilon_t$$

lateral strain

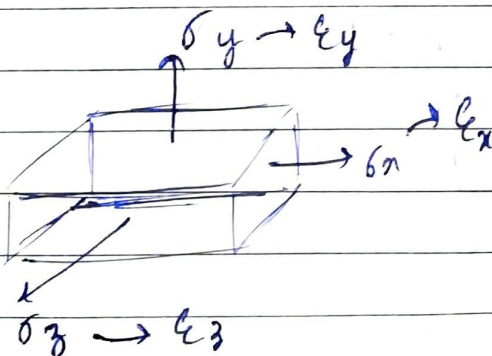
$$= \epsilon_l + 2\epsilon_t$$

$$\epsilon_v = \epsilon_l + 2\epsilon_t \quad \text{also} \quad \epsilon_t = \mu \epsilon_l$$

$$\epsilon_v = \epsilon_l - 2\mu \epsilon_l$$

$$= \epsilon_l (1 - 2\mu)$$

$$\epsilon_v = \frac{sl}{l} (1 - 2\mu)$$



Due to σ_x effect on y and z will be decreasing
 $-\mu \epsilon_x$ $-\mu \epsilon_x$

Due to σ_y effect on x & z will be decreasing
 $-\mu \epsilon_y$ $-\mu \epsilon_y$

Due to σ_z effect on x & y will be decreasing
 $-\mu \epsilon_z$ $-\mu \epsilon_z$

$$(\epsilon_x)_T = \epsilon_x - \mu \epsilon_y - \mu \epsilon_z$$

$$= \frac{\sigma_x}{E} - \mu \frac{\sigma_y}{E} - \mu \frac{\sigma_z}{E}$$

$$= \frac{1}{E} \left[\sigma_x - \mu \{ \sigma_y + \sigma_z \} \right]$$

$$(\epsilon_y)_T = \frac{1}{E} \left[\sigma_y - \mu \{ \sigma_x + \sigma_z \} \right]$$

$$(\epsilon_z)_T = \frac{1}{E} \left[\sigma_z - \mu \{ \sigma_x + \sigma_y \} \right]$$

$$\epsilon_v = \epsilon_x + \epsilon_y + \epsilon_z$$

$$= \frac{1}{E} \left[\sigma_x + \sigma_y + \sigma_z - 2\mu \{ \sigma_x + \sigma_y + \sigma_z \} \right]$$

$$= \frac{\sigma_x + \sigma_y + \sigma_z}{E} [1 - 2\mu]$$

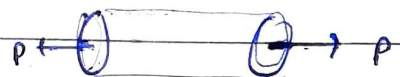
For cylindrical bars

$$V = \pi R^2 l$$

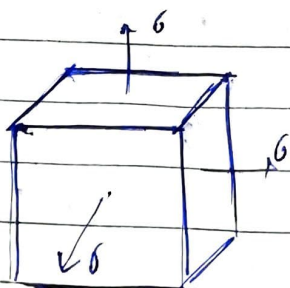
$$\ln V = \ln \pi + 2 \ln R + \ln l$$

$$\frac{dV}{V} = 0 + 2 \frac{dR}{R} + \frac{dl}{l}$$

$$\epsilon_v = 2 \epsilon_r + \epsilon_l$$

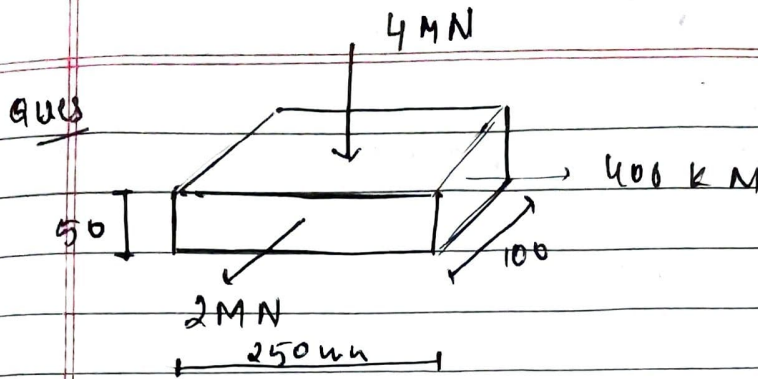


Bulk Modulus (K)



$$K = \frac{\sigma}{\epsilon_v} \quad \leftarrow \text{within elastic limit}$$

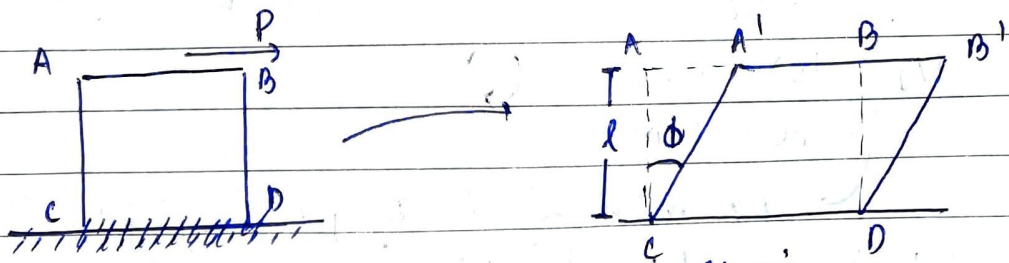
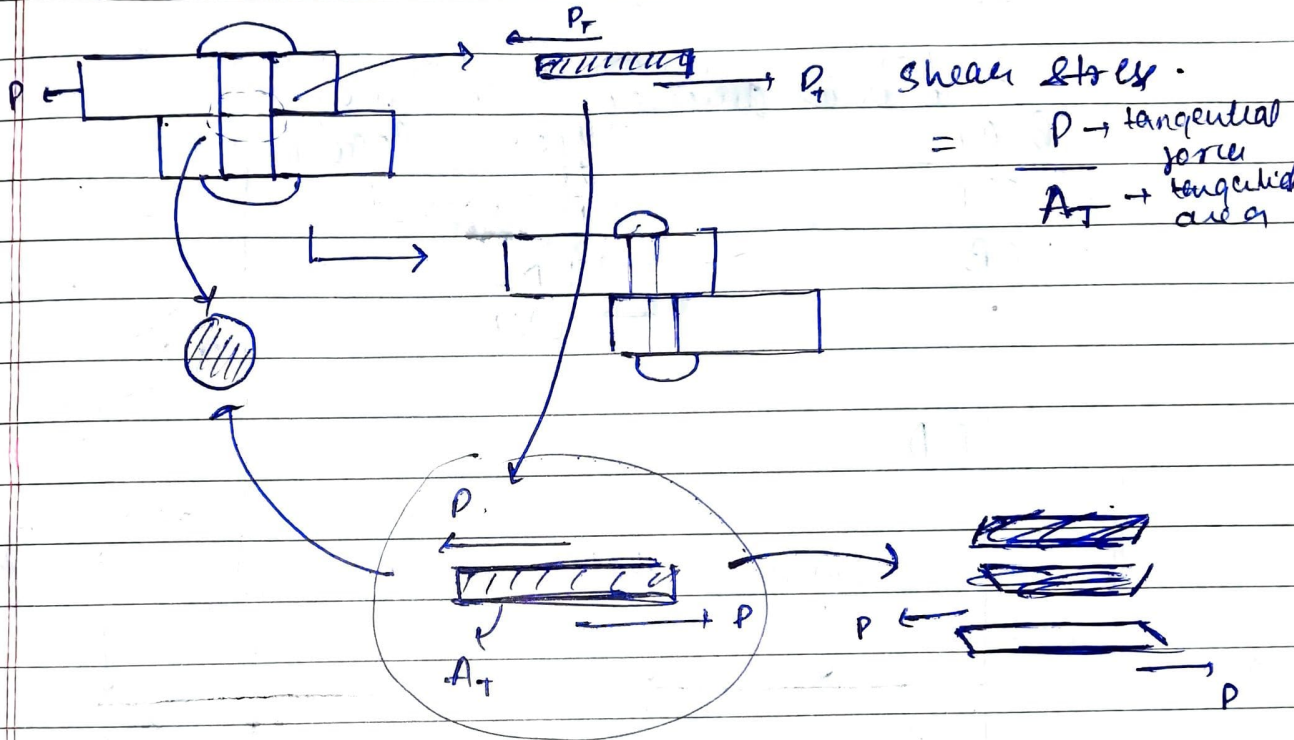
used when:-
equal intensity in all dirⁿ



$$\epsilon_z = -\epsilon_x - \mu \epsilon_y - \mu \epsilon_z$$

$$(\epsilon_z)_T = \frac{-\sigma_z}{E} - \frac{\mu}{E} [\sigma_x + \sigma_y]$$

Shear Stress



Shear ~~Strain~~ ^{Strain} $\phi = \frac{AA'}{l}$

within elastic limit :-

Shear Stress $\tau \propto \phi$

$\tau = G \phi$ $\Rightarrow$ $\frac{\tau}{\phi} = G$ $\rightarrow$ Elastic const

modulus of Rigidity

Poisson's ratio

Modulus of Elasticity /

Modulus of Rigidity

Young's Modulus

$$E = 3K(1 - 2\mu)$$

SFD / BMD

Shear force diagram Bending Moment diagram

→ whose 1 dimension is very large in comparison to other dimensions & can handle lateral load.

Beam

Types of loads

→ SSB

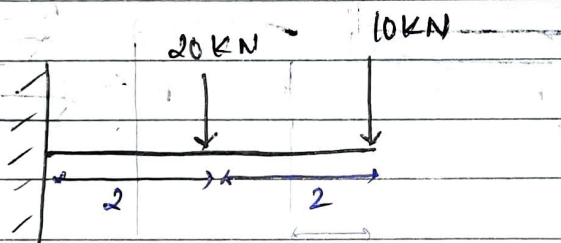
→ CB

→ Point load

→ UDL

→ UVL

SFD



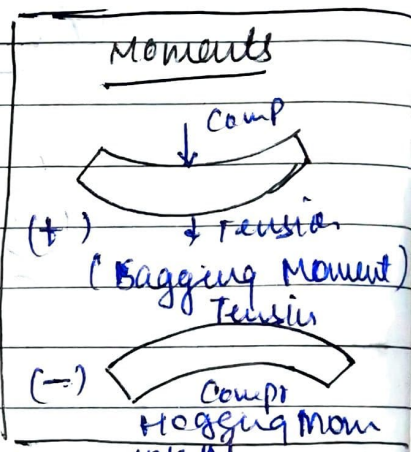
$$M_A = 10 \times 4 + 20 \times 2 \quad (\text{Clockwise})$$

$$= 80 \text{ kN-m}$$

→ Ye moment A pr lagega taaki vo 20 kN and 10 kN ki wajah se rotate na hojaye. Since A is fixed it will resist any type of movement.

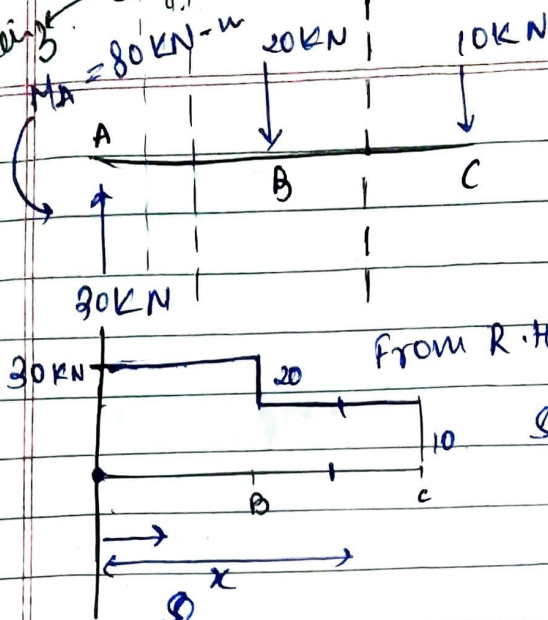
$$\rightarrow \sum F_x = 0$$

$$\rightarrow M_A = 0$$



Section 2
Section 1

Date _____
Page _____



For R.H.S

$\downarrow \rightarrow \oplus, \uparrow \rightarrow \ominus$

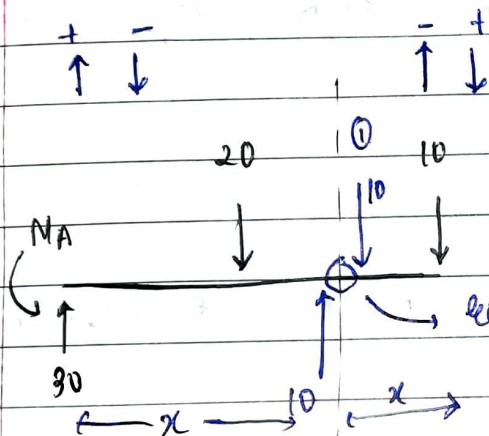
For L.H.S

$\downarrow \rightarrow \ominus, \uparrow \rightarrow \oplus$

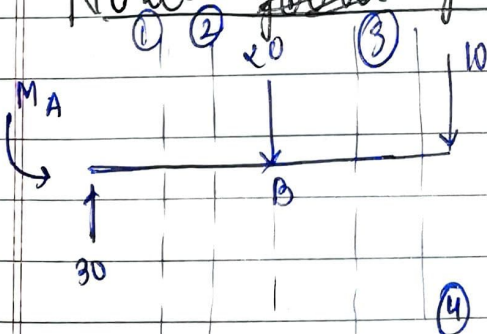
Section b/w B-C :-
 $SF = 10 \text{ kN}$

Section b/w A-B
 $SF = 10 + 20$

Now using SFD $\rightarrow$ SF at $x = 0.5$ -
 $(SF)_{0.5} = 30 \text{ kN}$



Now ~~from~~ for Section 3

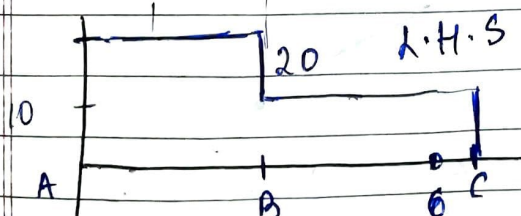


L.H.S

$\downarrow \uparrow$
 $\ominus \oplus$

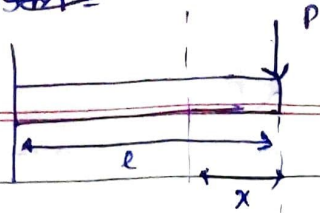
R.H.S

$\downarrow \uparrow$
 $\oplus \ominus$



$$30 - 20 - 10 = 0$$

~~B.M.F.~~

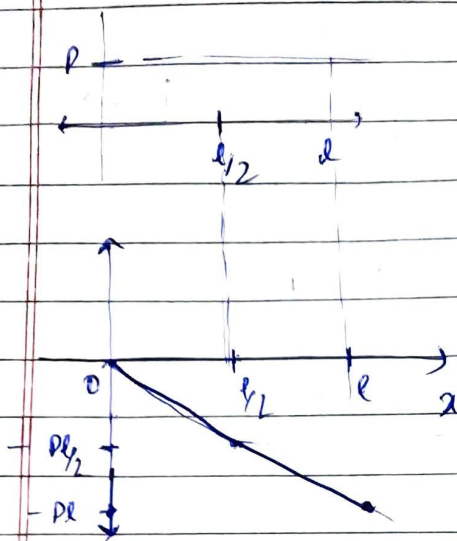


$$BM = -Px$$

$$x=0 \quad BM=0$$

$$x=l/2 \quad BM = -Pl/2$$

$$x=l \quad BM = -Pl$$



For RHS

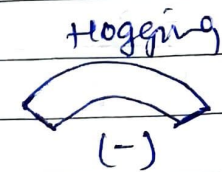
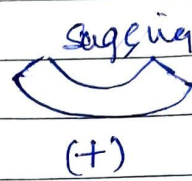
↓ (+)

↑ (-)

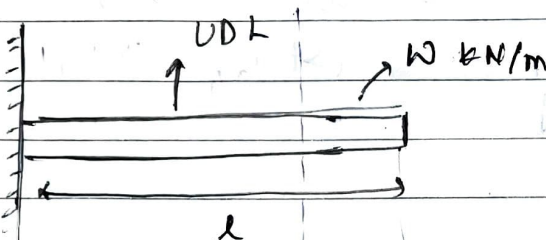
For LHS

↓ (-)

↑ (+)



Ques



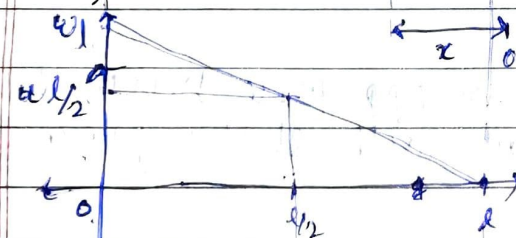
SFD

$$SF_x = w x$$

$$x=0 \quad SF=0$$

$$x=l/2 \quad SF = wl/2$$

$$x=l \quad SF = wl$$



BMD

$$BM_x = -w x \times \frac{x}{2}$$

$$= -\frac{w x^2}{2}$$

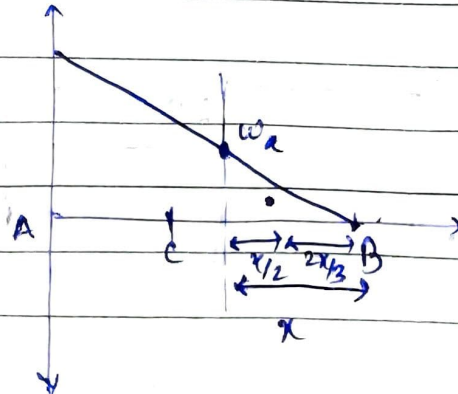
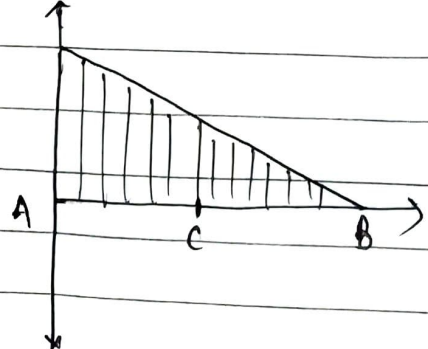
$$x=0 \quad BM=0$$

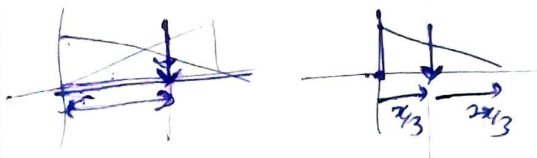
$$x=l/2 \quad BM = -\frac{wl^2}{8}$$

$$x=l \quad BM = -\frac{wl^2}{2}$$



Ques





$$\frac{wx}{l} = \frac{w}{l} \Rightarrow \boxed{w_x = \frac{wx}{l}}$$

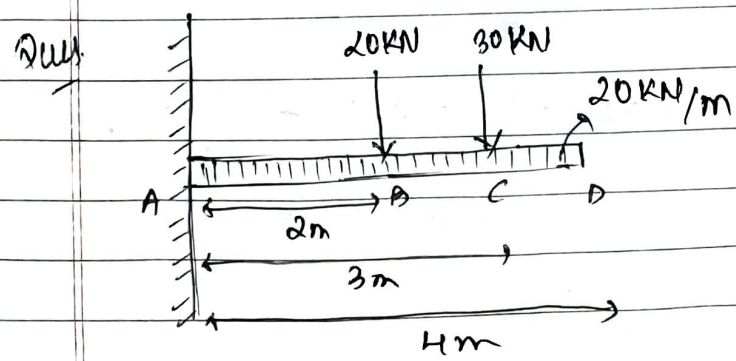
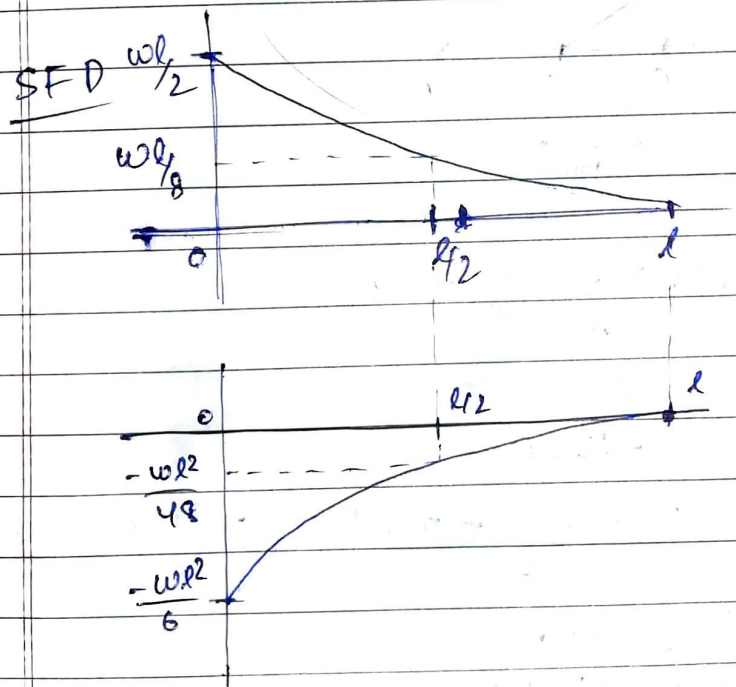
$$\frac{1}{2} \times w_x \times x = \frac{1}{2} \frac{wx^2}{l} \quad [\text{Total load acting at RHS}]$$

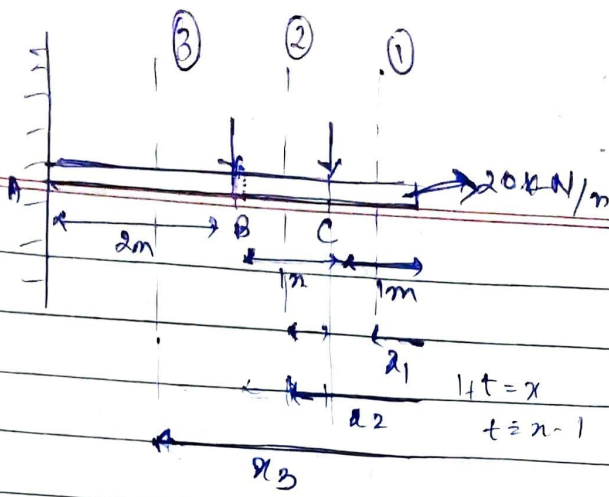
$$\text{SF.D} \Rightarrow SF_x = \frac{1}{2} \frac{wx^2}{l}$$

→ x = 0	SF = 0
→ x = l/2	SF = wl/8
→ x = l	SF = wl/2

$$\text{BMD} \Rightarrow BM_x = \frac{wx^2}{2l} \times \frac{x}{3} = -\frac{wx^3}{6l}$$

$x = 0 \quad BM = 0$
 $x = l/2 \quad BM = -\frac{wl^2}{48}$
 $x = l \quad BM = -\frac{wl^2}{6}$



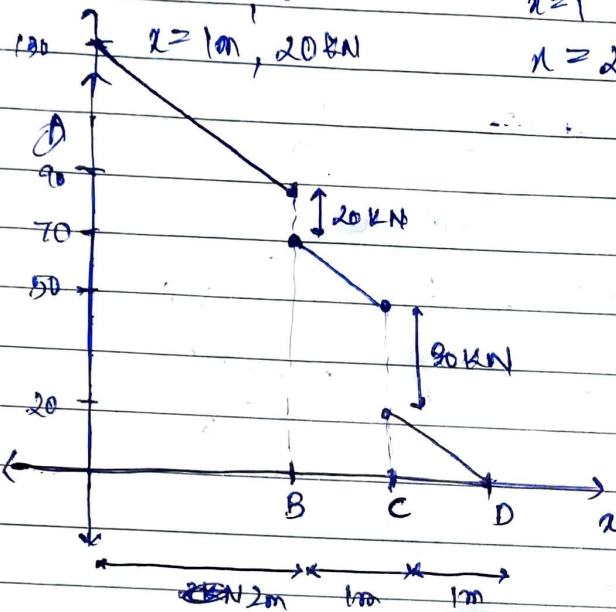


AB:- $SF_x = (20x + 30 + 20)$
 $x = 0.2 \text{ m}, 90 \text{ kN}$
 $x = 4 \text{ m}, (30 \text{ kN})$

SFD

CD:- $SF_x = 20x$ $x = 0, 0$ $x = 1 \text{ m}, 20 \text{ kN}$
 BC:- $SF_x = (20x + 30)$ $x = 1, 50 \text{ kN}$ $x = 2, 70 \text{ kN}$
 AB:- $SF =$

SFD



BMD

CD:- $BM_x = -20x \times \frac{x}{2} = -10x^2$

$x = 0, 0$
 $x = 1, -10 \text{ kN}$

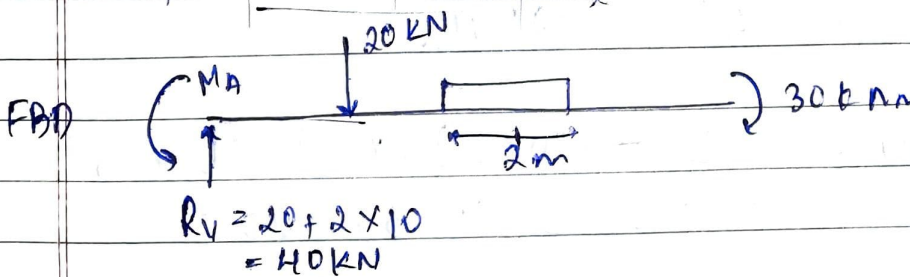
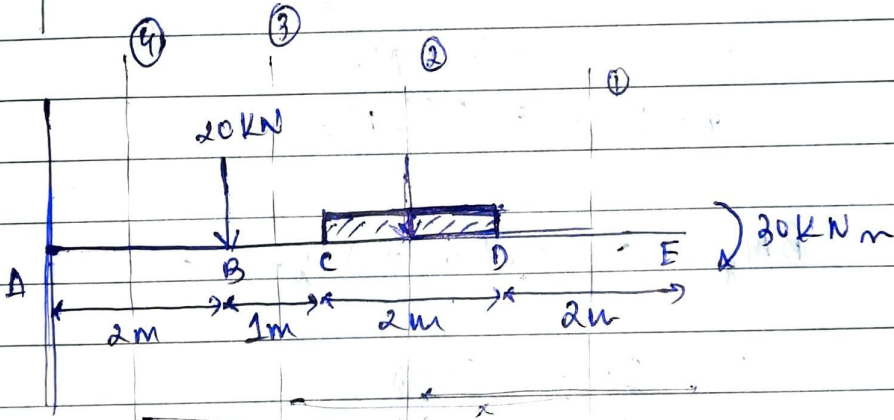
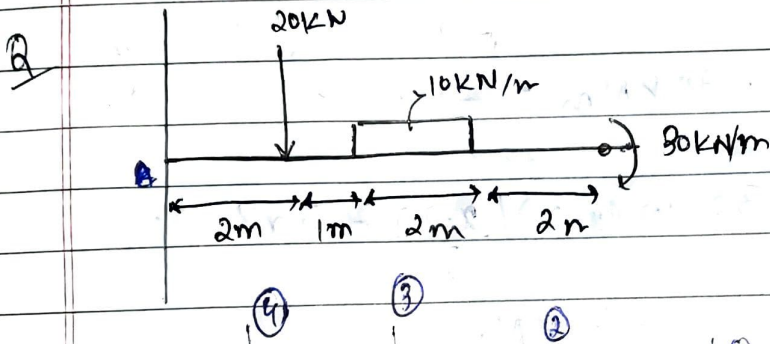
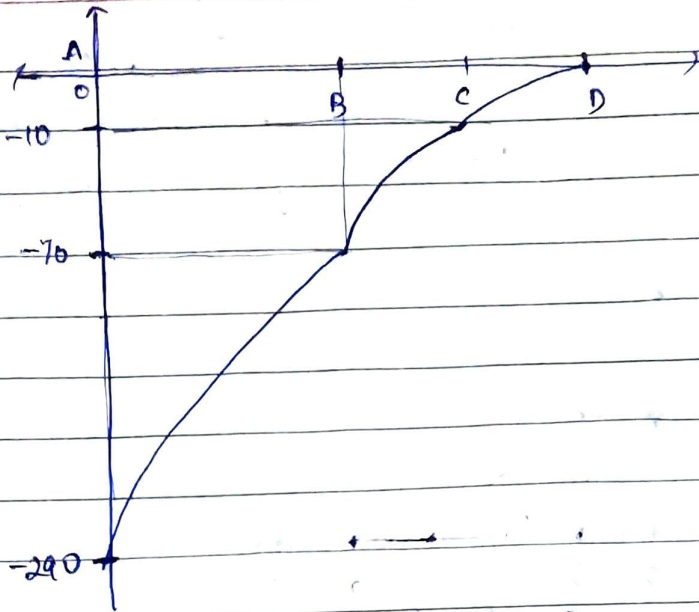
BC:- $BM_x = -20x \times \frac{x}{2} - 30(x-1) = (-10x^2 - 30x + 30)$

$x = 1, -10 \text{ kN}$
 $x = 2, -70 \text{ kN}$

AB:- $BM_x = -20x \times \frac{x}{2} - 30x(2-1) - 20(x-2)$

$= -10x^2 - 30x + 80 - 20x + 40$
 $= -10x^2 - 50x + 70$

$x = 2, -70 \text{ kN}$
 $x = 4, -290 \text{ kN}$



$$30 + 20 \times 2 + (10 \times 2 \times 4) + M_A = 0$$

$$M_A = -150 \text{ kNm} = 150 \text{ kNm (anti-clockwise)}$$

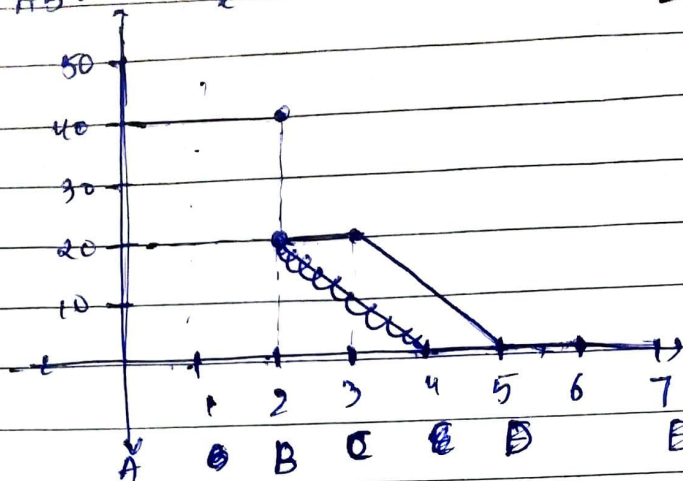
→ SFD :-

ED: $SF_x = 0$ — $\begin{cases} x=0, 0 \text{ kN} \\ x=2, 0 \text{ kN} \end{cases}$

CD: $SF_x = 10(x-2)$ — $\begin{cases} x=2, 0 \\ x=4, 20 \text{ kN} \end{cases}$

CB:- $SF_x = 10 \times 2 = 20 \text{ kN}$ $\left\{ \begin{array}{l} x=4, 20 \text{ kN} \\ x=5, 20 \text{ kN} \end{array} \right.$

AB:- $SF_x = 10 \times 2 + 20 = 40$ $\left\{ \begin{array}{l} x=5, 40 \text{ kN} \\ x=7, 40 \text{ kN} \end{array} \right.$



BMD:- ED:- $BM_x = -30 \text{ kNm}$ $\left\{ \begin{array}{l} x=0, -30 \text{ kNm} \\ x=2, -30 \text{ kNm} \end{array} \right.$

DC:- $BM_x = -30 - 10(x-2)\left(\frac{x-2}{2}\right) = 5x^2 - 20x + 50$

$x = 2, -30 \text{ kNm}$

$x = 4, -50 \text{ kNm}$

CB:- $BM_x = -30 - (10 \times 2)(x-3) = -30 - 20(x-3)$

$x = 4 \text{ m}, -50 \text{ kNm}$

$x = 5 \text{ m}, -70 \text{ kNm}$

BA:- $BM_x = -30 - (10 \times 2)(x-3) - 20(x-5)$
 $= -30 - 20(x-3) - 20(x-5)$

$x = 5 \text{ m}, -70 \text{ kNm}$

$x = 7 \text{ m}, -150 \text{ kNm}$

